

Yassine Nachit

Bio. I am a researcher in probability theory, focusing on generalizing some probability aspects from Gaussian to non-Gaussian settings. I am currently a postdoctoral researcher in Professor Ciprian Tudor group at the University of Lille, working mainly on Stein method and its applications to limit theorems and asymptotic independence. My interest in the Stein method developed while working with my former supervisor, Professor Ivan Nourdin, one of the creators of Malliavin-Stein method. I also developed a strong interest in machine learning during the postdoc at the University of Luxembourg, whose Department of Mathematics maintains an active research program in machine learning and regularly hosted lectures by leading experts e.g., Hanin Boris.

1- Research interests

(Malliavin) Stein method. Gaussian/non-Gaussian Probability Theory. Stochastic Analysis. Stochastic (partial) differential equations driven by Gaussian/non-Gaussian processes. Malliavin calculus. Sample path properties. local times for non-Gaussian processes. Besov regularity of stochastic processes. Gaussian density estimates.

2- Standard academic record

- PERSONAL INFORMATION

Name: Yassine NACHIT

Date of birth: 20/08/1991

Nationality: Moroccan

- Professional experience

2024 – to present **Postdoc at the University of Lille.** Supervisor: Ciprian Tudor

2023[†] – 2024 **Postdoc at the University of Luxembourg.** Supervisor: Ivan Nourdin

- Education

2016 – 2022[‡] **Ph.D. in Applied Mathematics**, Speciality Analysis and Probability, Cadi Ayyad University, Marrakesh, Morocco. Title: Local times and Besov regularity of the solution to the stochastic heat equation. Supervisor: Brahim Boufoussi

2014 – 2016 **Master's degree in Probability and Statistics**, Cadi Ayyad University, Marrakesh, Morocco. Master's Courses: probability, stochastic calculus, Malliavin calculus, etc. Master thesis : Local times for Gaussian processes . Supervisor: Brahim Boufoussi

2010 – 2013 **Bachelor of Fundamental Studies in Mathematics:** Cadi Ayyad University, Marrakesh, Morocco.

2009 – 2010 **Scientific Baccalaureate:** Marrakesh, Morocco.

3- Standard research record

- Preprints

[†] Visa processing period and administrative formalities (from March 2022 to February 2023)

[‡] Doctoral timeline includes two medical leaves, each lasting one year

Laurent Loosveldt, **Yassine Nachit**, Ivan Nourdin, Ciprian Tudor, “Malliavin smoothness of the Rosenblatt process” (2025). Preprint

Laurent Loosveldt, **Yassine Nachit**, Ivan Nourdin, Ciprian Tudor, “Absolute continuity of finite dimensional distributions of Hermite processes via Malliavin calculus” (2025). Preprint

This preprint is one of the most important works in my research: a two-year of effort—in this one and other related preprints that will be finalized soon—initiated during my postdoc at the university of Luxembourg. It develops in a creative way the first method that generalize the classical Gaussian approach for proving the absolute continuity of finite-dimensional distributions of Gaussian processes, e.g., fractional Brownian motion (Hermite processes for $q=1$), to a non-Gaussian framework, e.g., Hermite processes for $q>1$, via Malliavin calculus.

- *Publications*

Iulian Cîmpean, **Yassine Nachit**, and Ciprian Tudor, “Quartic variation of the solution to the semilinear stochastic heat equation: limit behavior and asymptotic independence with respect to the data”, to appear in Journal of Mathematical Analysis and Applications (2025).

This paper represents my first work on Stein method. It concerns using the classical Malliavin-Stein method introduced by Ivan Nourdin and Giovanni Peccati to derive the limit behavior of the quartic variation, i.e., the power variation of order four, with respect to the time variable of the solution to the semilinear stochastic heat equation with space-time white noise. Furthermore, in this article using a recent variant of Malliavin-Stein method introduced by Professor Ciprian Tudor, we analyze also the asymptotic independence between the quartic variation and the data which is used to construct it.

Brahim Boufoussi and **Yassine Nachit**, “On the uniform Besov regularity of local times of general processes”, to appear in Journal of Fractal Geometry (2025). arXiv:2110.10227

The main result of this preprint is to upgrade the uniform Hölder continuity for local times, in a Gaussian and non-Gaussian setting, to a uniform Besov regularity.

Brahim Boufoussi and **Yassine Nachit**, “Local times for systems of non-linear stochastic heat equations”, Stochastics and Partial Differential Equations: Analysis and Computations (2022). <https://doi.org/10.1007/s40072-021-00231-9>

This is one of the most important articles of my research: a two-and-a-half-year project that is among the first to extend local time theory from Gaussian to non-Gaussian setting via Malliavin calculus, establishing existence, joint continuity, and sharp Sobolev regularity for the local time of the (non-Gaussian) solution of the non-linear stochastic heat equation.

Brahim Boufoussi and **Yassine Nachit**, “On the Besov regularity of the bifractional Brownian motion”, Probability and Mathematical Statistics (2021), Vol. 41, Fasc. 2 (2021), pp. 303–320. <https://doi.org/10.37190/0208-4147.41.2.6>

In this article, we have generalized the work on the Besov regularity of the fractional Brownian motion (with stationary increments) to the bifractional Brownian motion (which does not have stationary increments), i.e., we have improved bifractional Brownian motion Hölder continuity results to a Besov regularity.

Brahim Boufoussi and **Yassine Nachit**, “On Besov regularity and local time of the stochastic heat equation”, Stochastics An International Journal of Probability and Stochastic Processes (2023), pp. 1–26. <https://doi.org/10.1080/17442508.2022.2164696>

In this article, we investigate the local time of the (Gaussian) linear stochastic heat equation—both in space and in time—and improve known sample-path Hölder continuity results to Besov-space regularity in each variable.

- Teaching activities

Summer Semester 2023: Lecturer for Mathematics Software, Bachelor's students, University of Luxembourg

Summer Semester 2023: Lecturer for Linear Algebra 2, Bachelor's students, University of Luxembourg

Winter Semester 2024: Lecturer for Optimization and Numerical Probability, Master in Data Science, University of Luxembourg

Winter Semester 2024: Tutor for Optimization and Numerical Probability, Master in Data Science, University of Luxembourg

Summer Semester 2024: Lecturer for Mathematics Software, Bachelor's students, University of Luxembourg

- Supervision experience

Department of Mathematics, University of Luxembourg, Supervisor of a Bachelor thesis on Illustration of the Central Limit Theorem and simulations of random variables in R

- Talks in Conferences

Plenary speaker at the Journées RT² ANAIS-MAIAGES 2025: Aléatoire et Fractales, Université Bretagne Sud, Vannes, France, April 2-4, 2025, Local times for systems of non-linear stochastic heat equations.

Densities for solutions of SDEs driven by non-Gaussian processes, Fractional Brownian Motion and its Applications, University of Luxembourg, Luxembourg, April 23-24, 2024.

Malliavin Calculus for SDEs driven by non-Gaussian processes, Stochastic Dynamics and Stochastic Equations, Bernoulli Center program at EPFL, Lausanne, Swiss, March 25-27, 2024.

Local times for systems of non-linear stochastic heat equations. Marrakesh International Workshop on Random Fractals and Markov Processes, Marrakesh, Morocco, November 21-26, 2022.

Local times for systems of non-linear stochastic heat equations. NSF/CBMS Conference: Gaussian Random Fields, Fractals, SPDEs, and Extremes, University of Alabama in Huntsville, USA, August 02-06, 2021.

Local time for stochastic heat equation. The 3rd International Conference on Research in Applied Mathematics and Computer Science (ICRAMCS 2021), Casablanca, Morocco, March 26-27, 2021.

On the local time of the solution of the non-linear stochastic heat equation. The 6th International Congress of the Moroccan Society of Applied Mathematics (SM2A), Beni Mellal, Morocco, November 07-09, 2019.

- Talks in Seminars

Densities for solutions of SDEs driven by non-Gaussian processes, Probability and Statistics Seminar, November 12, 2024, University of Lille, France.

Local times and Besov regularity of the solution to the stochastic heat equation, Probability and Statistics Seminar, October 20, 2022, Department of Mathematics, University of Luxembourg.

On the Besov regularity of some processes, Probability and Statistics Seminar, March 05, 2020, Faculty of Sciences Semlalia, Cadi Ayyad University, Marrakesh, Morocco.

Walsh's stochastic integral and stochastic heat equation, Probability and Statistics Seminar, December 01, 2016, Faculty of Sciences Semlalia, Cadi Ayyad University, Marrakesh, Morocco.

- Participation in Conferences

Luxembourg Workshop in Stochastic Analysis (2nd edition), University of Luxembourg, February 10-12, 2025, Luxembourg.

Luxembourg Workshop in Stochastic Analysis, University of Luxembourg, February 15-17, 2023, Luxembourg.

International conference on Malliavin calculus and related topics, University of Luxembourg, June 12-16, 2023, Luxembourg.

Probability and Statistics Day, National School of Applied Sciences, Cadi Ayyad University, September 27, 2016, Marrakesh, Morocco.

- Participation in Research Schools

Ecole Mathématique Africaine, EMA2015, Théorie générale des processus stochastiques et Introduction au calcul de Malliavin, Cadi Ayyad university, October 12-23, 2015, Marrakesh, Morocco.

Research School on Stochastic Analysis and its Applications (SSAA'2017), Faculty of sciences Semlalia, Cadi Ayyad university, April 3-13, 2017, Marrakesh, Morocco.

- Grants and Awards

Seal of Excellence (88%): MSCA-PF-2023 High Quality Project, 101155495 SPP HS-RRF, awarded by European Commission.

Seal of Excellence Laureate — Initiative d'excellence, Université de Lille (SoE-IE), 2025.

Recognition and support program for MSCA-PF Seal of Excellence holders at the university of Lille.

- Referee

Referee for the journal: Statistics & Probability Letters.

- Research Visits

October 2022 Research visit (1 week), Department of Mathematics, University of Luxembourg, Ivan Nourdin.

Research visit (26—28 January 2025), Department of Mathematics, University of Liège, Loosveldt Laurent

- Academic Projects

July 2016-February 2016	Local times for Gaussian processes: Existence of local times, Joint continuity of local times, Hausdorff dimension of level sets.
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- Mathematical skills

Probability and statistics, Analysis, Algebra, Financial mathematics, Stochastic analysis, Malliavin calculus.

- Programming Skills

C, C++, R, Scilab, LATEX, Python, SageMath

- Languages

Arabic, French, English